

# GDP 19 Electronics Test Plan

The purpose of this document is to detail the steps taken by GDP 19 to create and validate the electronic circuit of the canine elbow simulator.

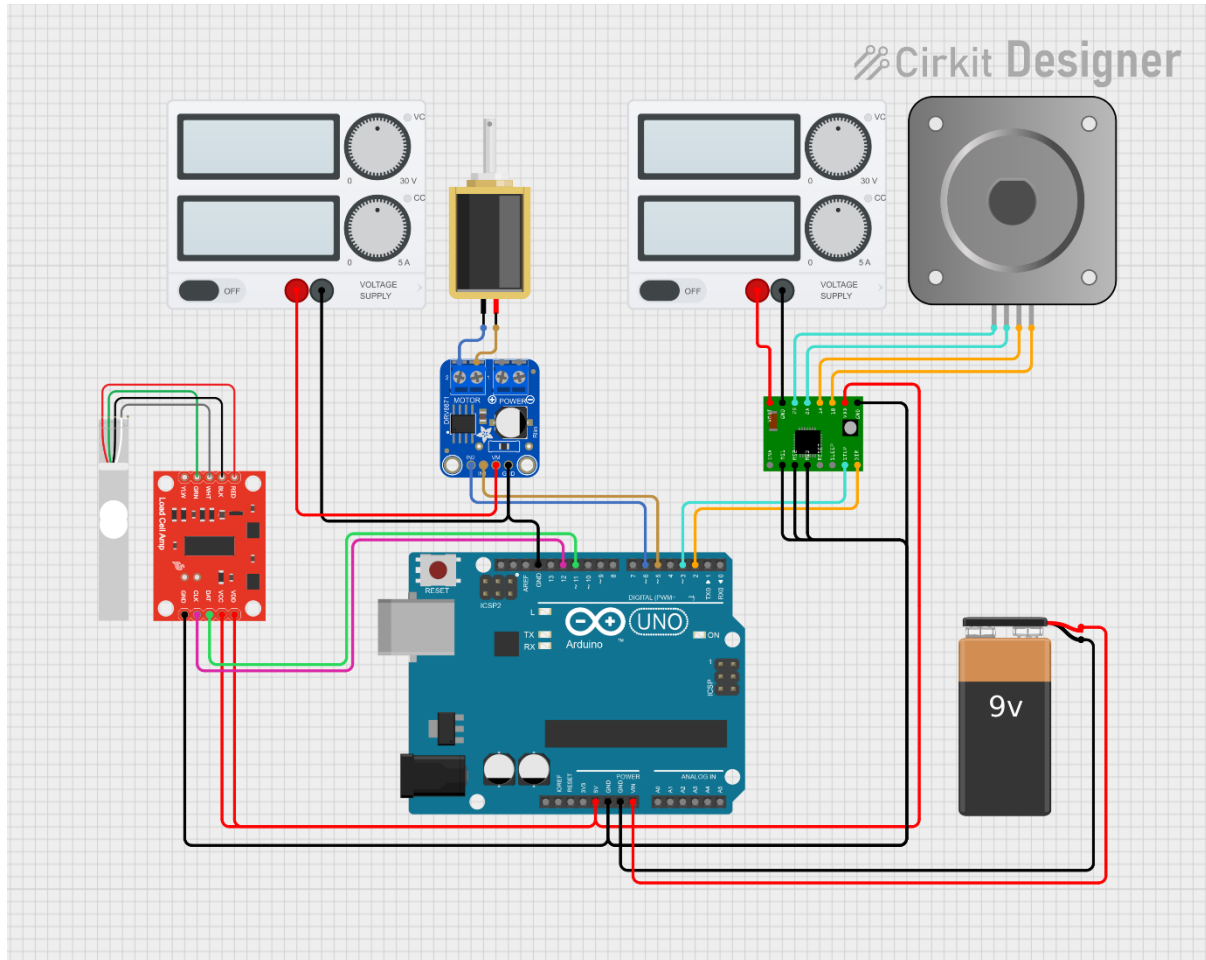


Figure 1: Electronics circuit diagram.

The circuit is comprised of:

- Stepper motor [Nema 17] system (for flexion/extension) controlled by motor driver A4988.
- Linear solenoid [Ledex 5SF] system (for force application) controlled by motor driver DRV8871.
- Load cell [PSD-S1] connected with load cell amplifier HX711.
- ARDUINO UNO R3 microcontroller interfaced to computer with serial communication.

## 1. Breakout Boards

The motor drivers and load cell amplifier are used as breakout boards for prototyping. Male pin headers are soldered to the breakout board to allow for breadboard prototypes to be configured easily. Terminal blocks are also used for the motor drivers for power delivery.

- Solder male pin headers.
- Solder terminal blocks.

## 2. Stepper Motor

The stepper motor, Nema 17, is used with a motor driver, A4988, for basic control. The A4988 is excessive for the application of 70° rotation at 1+ Hz, though due to its availability it's simple to implement.

Issues may be caused from the stepper motor missing steps, leading to a large error over hours of rig operation. This could cause the bones to slip and cause damage to the rig. The motor may miss steps due to the high torque required of the changing directions due to flexion extension. Furthermore, torque may be 'lost' to the flexible shaft coupler.

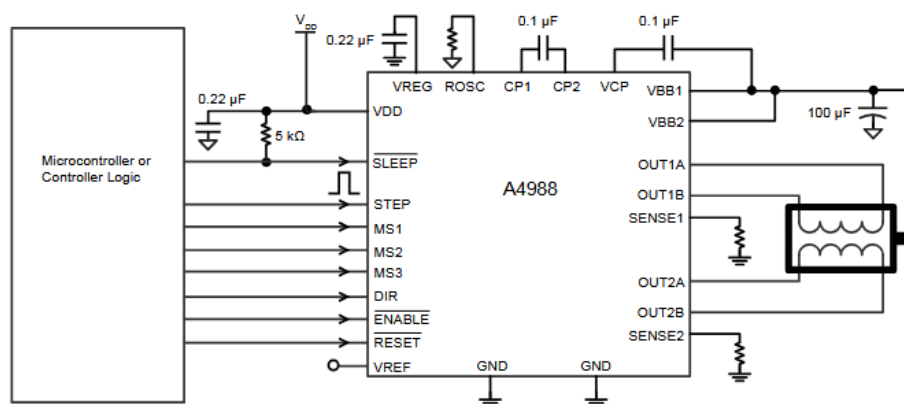


Figure 2: A4988 Application Diagram. Passive components are internal to the breakout board.

- Connect Nema 17 to A4988 with power supply.
- Test 70° of articulation at 1 Hz.
- Test 70° of articulation at 2.2 Hz.
- Measure impact of connection to bottom sub-assembly (with flexible shaft coupler).

Testing notes:

- A4988 requires external cooling for operating over long periods when connected to the bottom sub-assembly. Fix using either a heatsink or different, more efficient driver (DRV8825).
- Larger stepper motor (holding torque) required to increase rotational acceleration.

3. Linear Solenoid

The linear solenoid, Ledex 5SF, is used with a motor driver, DRV8871, for actuation with pulse width modulation (PWM). Linear solenoids are binary actuators, meaning they are either fully open or full closed, which means PWM is required to provide varying forces.

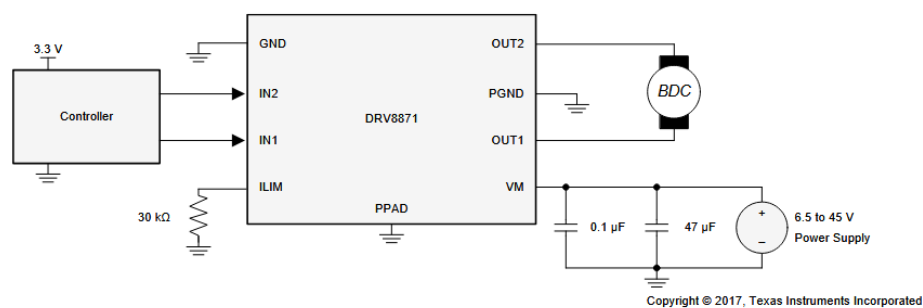


Figure 3: DRV8871 Application Diagram.

|                                                                                        |                                                                |                       |                         |              |              |              |              |
|----------------------------------------------------------------------------------------|----------------------------------------------------------------|-----------------------|-------------------------|--------------|--------------|--------------|--------------|
| LINEAR Low Profile                                                                     | Short Stroke, Flat Face<br>Part Number: 129450-0XX             |                       |                         |              |              |              |              |
|                                                                                        | Performance                                                    |                       |                         |              |              |              |              |
|                                                                                        | Maximum Duty Cycle                                             | 100%                  | 50%                     | 25%          | 10%          |              |              |
|                                                                                        | Maximum ON Time (sec)<br>when pulsed continuously <sup>1</sup> | ∞                     | 100                     | 36           | 10           |              |              |
|                                                                                        | Maximum ON Time (sec)<br>for single pulse <sup>2</sup>         | ∞                     | 160                     | 44           | 13           |              |              |
|                                                                                        | Watts (@ 20°C)                                                 | 21                    | 42                      | 84           | 210          |              |              |
|                                                                                        | Ampere Turns (@ 20°C)                                          | 860                   | 1220                    | 1720         | 2730         |              |              |
|                                                                                        | Coil Data                                                      |                       |                         |              |              |              |              |
|                                                                                        | avg<br>(0XX) <sup>3</sup>                                      | Resistance<br>(@20°C) | #<br>Turns <sup>4</sup> | VDC<br>(Nom) | VDC<br>(Nom) | VDC<br>(Nom) | VDC<br>(Nom) |
|                                                                                        | 23                                                             | 2.03                  | 288                     | 6.1          | 8.6          | 12.1         | 19.2         |
|     | 24                                                             | 3.20                  | 360                     | 7.6          | 10.8         | 15.3         | 24.0         |
|                                                                                        | 25                                                             | 4.91                  | 440                     | 9.6          | 13.6         | 19.2         | 31.0         |
|                                                                                        | 26                                                             | 7.72                  | 550                     | 12.1         | 17.1         | 24.0         | 38.0         |
|                                                                                        | 27                                                             | 11.12                 | 636                     | 15.0         | 21.0         | 30.0         | 48.0         |
|                                                                                        | 28                                                             | 18.79                 | 840                     | 19.2         | 27.0         | 39.0         | 61.0         |
|                                                                                        | 29                                                             | 30.48                 | 1088                    | 24.0         | 34.0         | 48.0         | 77.0         |
|                                                                                        | 30                                                             | 44.86                 | 1275                    | 30.0         | 43.0         | 61.0         | 96.0         |
|                                                                                        | 31                                                             | 70.90                 | 1596                    | 38.0         | 54.0         | 76.0         | 121.0        |
|                                                                                        | 32                                                             | 109.00                | 1974                    | 47.0         | 67.0         | 95.0         | 150.0        |
|                                                                                        | 33                                                             | 175.00                | 2496                    | 60.0         | 86.0         | 121.0        | 192.0        |
| <sup>1</sup> Continuously pulsed at stated watts and duty cycle                        |                                                                |                       |                         |              |              |              |              |
| <sup>2</sup> Single pulse at stated watts (with coil at ambient room temperature 20°C) |                                                                |                       |                         |              |              |              |              |
| <sup>3</sup> Other coil avg sizes available — please consult factory                   |                                                                |                       |                         |              |              |              |              |
| <sup>4</sup> Reference number of turns                                                 |                                                                |                       |                         |              |              |              |              |

Figure 4: Ledex 5SF Datasheet. avg-029 used for project.

- Connect Ledex 5SF to DRV8871 with passive components and power supply.
- Test full-stroke actuation.
- Implement PWM for half-power actuation.

#### Testing Notes:

- DRV8871 capable of 45 V, however tabletop lab power supplies can only provide 30 V. Due to ohmic relationship, linear solenoid is capped at 24 V 1.5 A, resulting in low force output. Fix with lower awg linear solenoid or higher voltage power supply.
- PWM implementation is non-linear leading to dead-zone between PWM 0 → PWM 100. Remap with ARDUINO.
- Solenoid force output very dependent on spacing between pin and housing, mitigated with 3D printed horseshoe shim. Future work should look at formal ways of calibrating the linear solenoid pin.

## 4. Load Cell

The load cell, PSD-S1 100kg, is used in conjunction with a load cell amplifier, HX711, to create readable data for the control system. The PSD-S1 uses an excitation voltage of 9~12 V DC, however the HX711 uses a supply voltage of 2.7~5.5 V DC. The load cell can still be run on this lower voltage however will provide a weaker signal strength, increasing the impact of noise.

Noise is induced in multiple ways: stability of power supply, parasitic electromagnetic fields (linear solenoid), capacitive coupling of power cables. To mitigate this, shielded wires should be used to prevent interference from other electromagnetic fields. The system should be grounded to a single point to prevent a ground loop and minimal cable length should be used. Additionally, noise can be filtered with the microcontroller, implemented with a moving window filter.

The load cell requires calibration using an Instron electromechanical test machine at the Testing and Structures Research Laboratory. Given the load cell capacity of 100 kg, testing is applied in increments of 200 N, up to 800N and validated at 950N. Calibration involves adjusting the calibration factor of the HX711, detailed on example scripts in the HX711.h ARDUINO library.

A calibration factor of -43675 was found to best represent kilograms.

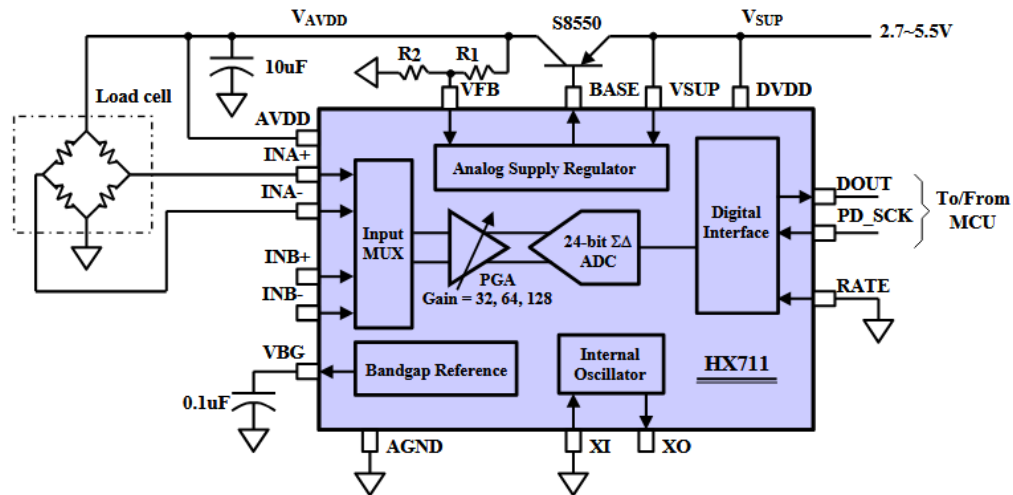


Figure 5: HX711 Application Diagram. Passive components are internal to the breakout board.

- Connect PSD-S1 with HX711 with 5 V.
- Implement HX711 ARDUINO library.
- Calibrate the load cell using Instron electrodynamic test machine.
- Reduce noise using moving window filtering.

#### Testing notes:

- Calibration factor of -43675 was found to be suitable.
- Sample rate needs to be controlled for PID implementation.
- To further reduce noise, send the average of 3 readings to the window filter.
- Window filter causes latency.

## 5. Compliance Test

The stiffness of the load cell assembly should be investigated to determine internal compliance. Given the custom parts are made from Al-6082 T6, the compliance is dominated by the load cell. Therefore, during calibration, a control sample of extremely high stiffness should also be tested at the same forces to find the difference in deflection.

- Test extremely stiff sample to determine compliance of the load cell.

#### Testing notes:

- Load cell was found to have minimal compliance.  $83 \mu\text{m}$  at  $0.95 \text{ kN}$ , compliance of  $8.73 \times 10^{-8} \text{ m/N}$ .

## 6. Control System

A control system is required to provide feedback to the ARDUNIO on the rate of PWM applied to the linear solenoid, hence controlling force.

A PID controller is to be implemented.

- Proportional term required to correct error. Unable to fully correct steady-state error.
- Integral term used to eliminate steady state error. Can cause slower responses.
- Derivative term to achieve faster response times, critical for high frequency application.

Derivative is sensitive to noise, therefore signal processing needs to be applied, such as a Kalman filter or moving window filter. Ziegler-Nichols method can be used to tune the PID controller given the system is difficult to model. Then, gain scheduling control will be implemented to tune gains for each movement type due to the non-linearity of the solenoid spring / load cell.

When testing the PID controller, care must be taken to prevent the derivative term from sending dangerous feedback to the ARDUINO which may cause the solenoid to be supplied excess current. The simplest solution will be to operate with a power supply with a current limit and apply the current limit according to the maximum wattage of the relevant duty cycle. Integral windup needs to be avoided when using a current limit, controlled via the ARDUINO scripts.

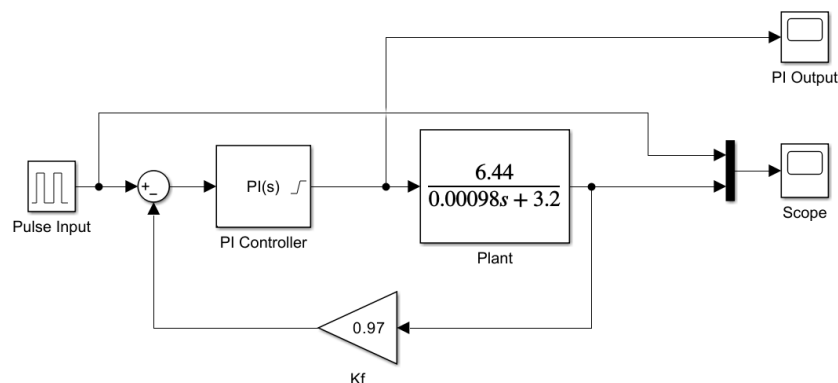


Figure 6: Control system experimental values.

Gain scheduling control is explored to provide different gains for the different movement cycles. Due to the compliance of the force actuator assembly, the gains tuned at low forces won't be as effective at higher forces. Therefore, the system can be tuned for multiple different movement types.

- Follow Ziegler-Nichols method of implementing PID controller.

- Start with stand force (120 N constant), then walk (240 N 1 Hz), then run (370 N 2 Hz).
- Use gain scheduling control to apply PID control to each movement cycle.

## 7. Python Integration

Python is used for the graphical user interface

- Establish serial communication with Python.
- Set up routines with ARDUINO.
- Integrate moving dot to demonstrate current progress of program.
- Implement force feedback on Python pilot.
- Implement an initialisation routine.

## 8. Test Plan

The following test plan is used to individually test each component and ensure the system can operate.

- Test rotation without load
- Test force with stiff metal rod
- Test feedback with stiff metal rod

Then, after potting a set of bones:

- Test force with bones
- Test rotation with bones
- Test feedback with bones

Then, after verifying:

- Test phase of the force and rotation from ARDUINO
- Test from Python, receiving feedback

## 9. Future Works

- Formal integration of PID controller to ARDUINO script.
- Non-blocking delays in ARDUINO script
- Feedback of stepper position using encoders → diametric magnet + rotary hall effect encoder.